

Fractions, Decimals and Percentages — Paper D

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- Give fractions in their simplest form unless the question says otherwise.
- The mark for each question is shown in square brackets on the right.
- Two questions on this paper have no obvious first step. If one stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – PROFIT AND LOSS

- 1.** A trader buys an item for 480 rupees and sells it for 600 rupees. Calculate the percentage profit. [3]

- 2.** A shopkeeper sells a bicycle at a 20% loss, receiving 640 rupees for it. Find the price he originally paid. [3]

A 20% loss means he received 80% of what he paid. That is the same shape of problem as a reverse percentage.

SECTION 2 – FRACTIONS OF WHAT IS LEFT

- 3.** Anil spends $\frac{1}{3}$ of his money on books, and then $\frac{1}{4}$ of what remains on food. He is left with 900 rupees.

- (a)** What fraction of his original money does he have left? [2]

(b) How much money did he start with?

[2]

4. A tank is $\frac{3}{5}$ full. After 240 litres are removed, it is $\frac{1}{3}$ full. Find the total capacity of the tank. [4]

SECTION 3 – COMPARISONS AND TRAPS

5. A shop offers two deals on an item priced at 600 rupees. [4]

Deal 1: 20% off, then a further 10% off.

Deal 2: 25% off, then a further 5% off.

Work out the final price under each deal, and state which is better and by how much.

6. The price of an item rises by 25%. By what percentage must the new price now fall to bring it back to exactly the original price? [4]

The answer is not 25%. Work with a starting value of 100 if that helps you see it.

- 7.** In a school of 500 students, $\frac{3}{5}$ are girls. 25% of the girls and 40% of the boys walk to school. How many students walk to school in total? [4]

- 8.** A student has written two answers in her book, and both are wrong. [4]

(i) "0.9 is smaller than 0.15, because 15 is bigger than 9."

(ii) " $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$ "

For each one, explain the mistake and give the correct answer.

End of paper. Check every answer has a unit or a form the question actually asked for.